Program

```
 #include <stdio.h>
#define MAX 50
int main()
{
    int graph[MAX][MAX];
   int visited[MAX];
   int queue[MAX];
   int n, i, j;
   int front = 0, rear = 0;
   int start, current;
  // Read number of vertices
  printf("Enter number of vertices: ");
  scanf("%d", &n);
  // Read adjacency matrix
  printf("Enter the adjacency matrix:\n");
  for (i = 0; i < n; i++)
  {
     for (j = 0; j < n; j++)
     {
       scanf("%d", &graph[i][j]);
     }
  }
  // Initialize visited array
  for (i = 0; i < n; i++)
  {
   visited[i] = 0;
  }
  // Read starting vertex
  printf("Enter starting vertex: ");
  scanf("%d", &start);
  // Insert starting vertex into queue
  visited[start] = 1;
  queue[rear] = start;
  rear++;
  printf("BFS Traversal: ");
  // BFS traversal
  while (front < rear)
```

```
  {
     // Remove vertex from queue
    current = queue[front];
    front++;
    printf("%d ", current);
    // Find adjacent vertices
   for (i = 0; i < n; i++)
   {
      if (graph[current][i] == 1 && visited[i] == 0)
      {
         visited[i] = 1;
        queue[rear] = i;
        rear++;
      }
    }
  }
  return 0;
}
```

Sample Input

Enter number of vertices: 5
Enter the adjacency matrix:
0 1 1 1 0
1 0 1 0 0
1 1 0 0 1
1 0 0 0 0
0 0 1 0 0
Enter starting vertex: 0

Output

BFS Traversal: 0 1 2 3 4